Centers for Medicare & Medicaid Services

CMS Target Life Cycle (TLC)

4Innovation (4i) Command Line Interface (CLI)

User Guide

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Notices and Disclaimers

When a PDF, this document has been tested and is accessible with JAWS 11.0 or higher.

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1 Overview

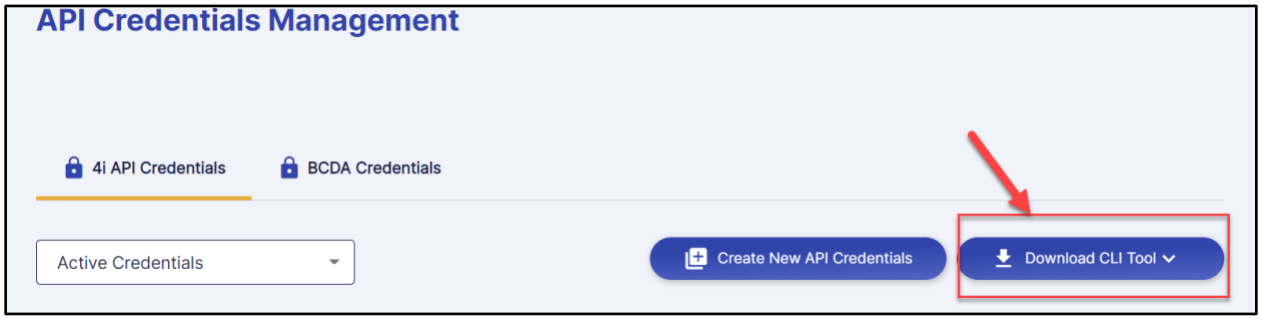
The 4Innovation (4i) Command Line Interface (CLI) is a tool that provides easy access to data within the 4i system. It utilizes the Application Programming Interface (API) Key management section of 4i to provide command line access to the Data Hub workspace. This overview will walk through the basics of the CLI, including setup and basic usage.

2 CLI Setup

2.1 Download

To use the 4i CLI, you must first download the tool from the 4i system (see Figure 1 below). After having downloaded the CLI, it is available for immediate use.

Figure 1: API Credentials Management in 4i



Note: The location of the CLI on your computer is important. Be sure to put it in a location that will be easy to access and easy to remember. By default, when a file is downloaded from a website, it will be available in the downloads folder of your computer. It is recommended to move it from this folder to a more common one for easier access.

2.2 Configuration

Before the CLI can be used to access data within 4i, it must be configured with API Client credentials. These must be generated from the 4i API Credential Management workspace under the 4i API Credentials tab.

Please reference the "*API Credentials Management Tip Sheet*" in your 4i Knowledge Library workspace for additional information on how to create API Key.

Opening a Command Prompt in a Specific Folder (Windows)

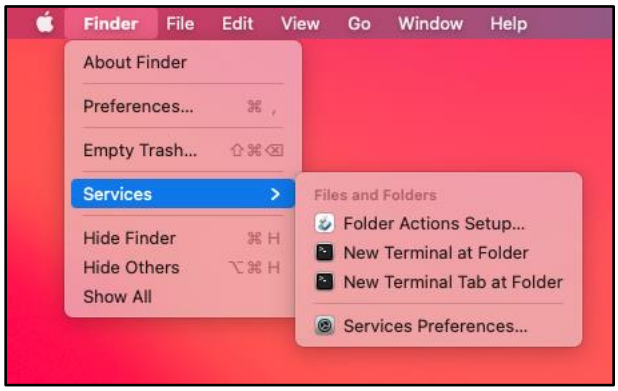
Using the File Explorer tool on your computer, navigate to the folder where the CLI has been downloaded, then click the address bar and type **"cmd"** and hit Enter on your keyboard.

A full walkthrough of this process is provided in the “How Do I Open a Terminal Window in a Folder” section at the following link: https://www.lifewire.com/open-command-prompt-in-a-folder-5185505

Opening a Terminal in a Specific Folder (Mac)

Open a Finder window on your Mac and select (but do not navigate to) the folder containing the downloaded CLI. Then in the top menu, select **“Finder” > “Services” > “New Terminal at Folder.”**

Figure 2: “Finder” > “Services” > “New Terminal at Folder”



Configuring Access Keys

The first thing required to use the 4i CLI is to configure your access credentials. This is done by running the following command:

> ./4icli configure

Figure 3: > ./4icli configure

This command will prompt you to enter your API Client Key and API Client Secret. After you have done this, you are ready to use the CLI.

Note: After you have configured your CLI, if you move the executable to a different folder you will need to run the 4icli configure command again.

Delinking API Keys for Terminated Contact(s)

If a Contact is terminated from an entity:

- If a terminated Contact has an active API Key for one (1) entity, that key will automatically be revoked. To see the revoked key for your entity, click on the “Revoked Key” drop-down in the “API Credentials” page.

- If a terminated Contact has an active API Key for multiple entities, only the entity associated with the terminated Contact will have its API key revoked. The terminated Contact will still see and have an active API Key for the remaining entities where they are an active Contact.

Note: API Keys where an entity has been delinked will no longer retrieve information from the delinked entity when using the CLI tool to query the 4i system.

3 Using the CLI

The 4i CLI is run from a terminal or command prompt and meant to be used programmatically.

To view a list of commands, you can run **4icli -h**. This will show you a list of commands.

The 4i CLI currently supports accessing data from the 4i Data Hub. To view a list of available commands with Data Hub run: **4icli datahub -h**. This will show a list of options for accessing Data Hub.

3.1 Usage

To run any command from the 4i CLI in your command prompt or terminal, simply enter “4icli” followed by the command you wish to run, for instance:

> ./4icli datahub -h

Or

> ./4icli datahub –help

This will show you all available commands for Data Hub within the CLI. More command options are listed in the following table.

Table 1: Options

Command	Description
-a, --apm=apm	Alternative Payment Model (APM) Entity ID
-c, --category=(Beneficiary List,CCLF,Reports,Monthly Exclusion Files) [default: CCLF]	Data Hub category to download files
-d, --download	Download files
-f, --file=file [default: 113]	File Type code to download a single file type
-h, --help	Show CLI help
-l, --list	List Data Hub folders and file types
-v, --view	View list of files

Command	Description
-y, --year=year [default: 2020]	Performance Year
--createdAfter=createdAfter	Get files created after the provided date. Example: --createdAfter=2021-05-01
--createdBetween=createdBetween	Get files created between the provided dates. Example: --createdBetween=2021-05-01,2021-06-01
--createdWithinLastMonth	Get files created within the last month.
--createdWithinLastWeek	Get files created within the last week.
--updatedAfter=updatedAfter	Get files updated after the provided date. Example: --updatedAfter=2021-05-01
--updatedBetween=updatedBetween	Get files updated between the provided dates. Example: --updatedBetween=2021-05-01,2021-06-01

Note: *Unless you are running the –help command or the –list command, specifying an Entity ID with -a or –apm is required.*

3.2 Example Commands

To view all categories within Data Hub, use the following command:

> ./4icli datahub –list

To view all Claim and Claim Line Feed (CCLF) files in a category for an Alternative Payment Model (APM), use the following command:

> ./4icli datahub --apm=C0001 --category=CCLF --view

To download all CCLF files in a category for an APM, use the following command:

> ./4icli datahub --apm=C0001 --category=CCLF --download

To download all CCLF files created in the past month for an APM, use the following command:

> ./4icli datahub --apm=C0001 --category=CCLF --createdWithinLastMonth --download

To download all CCLF files created in the past week for an APM, use the following command:

> ./4icli datahub --apm=C0001 --category=CCLF --createdWithinLastWeek –download

4 Rotate API Credentials

4i API credentials expire 90 days after they are created. In order to keep API credentials active past the 90 day expiration, the key must be rotated. Rotating the API credentials will assign a new Client Secret (private key), while the Client Key (public key) will remain unchanged. 4i API credentials can be rotated through the CLI by performing the following command:

> ./4icli rotate

After running the above command, you will be prompted with a confirmation message. To proceed, enter ‘y’ to proceed or ‘n’ to cancel and press enter. After selecting ‘y’, the CLI tool will perform the key rotation and automatically update the CLI tool configuration. The Client Key and Client Secret will also be displayed on the screen to copy and store somewhere safe. At this point the API credential rotation process is complete and the expiration date is renewed to 90 days.

5 Frequently Asked Questions

5.1 When I double click the CLI, nothing happens

The 4i CLI is an executable that must be run from the command prompt or terminal. Please refer to the configuration section (2.2) in this user guide for how to open a command prompt or terminal to run commands against the CLI.

5.2 When I run the 4icli … command, I get an error “No such file or directory”

Please try running the command again but include .”/” (a period and forward slash) at the beginning of the command. For example:

> ./4icli datahub configure

In order to run a command from a terminal or command prompt without .”/” at the beginning, you must have that directory in your path.

In order to move into a directory from within the terminal, you can use the following cd command:

> cd [Name Of Directory]

Or if you need to move up a directory, use:

> cd ..

Acronyms

Term	Definition
4i	4Innovation
API	Application Programming Interface
APM	Alternative Payment Model
CCLF	Claim and Claim Line Feed
CLI	Command Line Interface

Revision History

Version	Date	Organization/POC	Description of Changes
0.1	12/06/2021	Softrams/ Christopher Hand/ Taiwo Abimbola	Initial draft
1.0	12/10/2021	Softrams/ Kristina Kriss	Final baseline copy.
1.1	08/28/2023	Softrams/ Ryan Hill	Added section “Delinking API Keys for Terminated Contacts”
2.0	08/31/2023	Softrams/ Kristina Kriss	Final baseline copy.
2.1	05/09/2024	Softrams/ Micah Courey	Added instructions for rotate